

16th Mar 2026 | DA6360

⇒ Momentum (Heavy Ball) memory of past gradients

$$v_{t+1} = \beta v_t - \nabla f(x_t)$$

$$x_{t+1} = x_t + \alpha \cdot v_{t+1} \leftarrow \text{velocity-based push}$$

$$x_{t+1} = x_t + \beta(x_t - x_{t-1}) - \alpha \nabla f(x_t)$$

⇒ Nesterov

push a bit in the same "direction" as the past

$$y_{t+1} = x_t + \beta(x_t - x_{t-1})$$

$$x_{t+1} = x_t - \alpha \nabla f(y_{t+1})$$

gradient is calculated at pushed pt.

works better because we are doing a lookahead (futuistic gradient)

$$x_{t+1} = x_t + \beta(x_t - x_{t-1}) - \alpha \nabla f(x_t + \beta(x_t - x_{t-1}))$$

does a better estimate

⇒ Newton Method

perform second order approximation
around pt. x_t

$$f(x) \approx \hat{f}_t(x) = f(x_t) + \langle x - x_t, \nabla f(x_t) \rangle + \frac{1}{2} (x - x_t)^T \nabla^2 f(x_t) (x - x_t)$$

solve for argmin $\hat{f}_t(x)$

$$\nabla_x \hat{f}_t(x_t) = \nabla_x f(x_t) + \nabla_x^2 f(x_t) (x - x_t)$$

we know that,

$$\nabla_x \hat{f}_t(x_{t+1}) = 0$$

$$\Rightarrow \nabla_x \hat{f}_t(x_{t+1}) = \nabla_x f(x_t) + \nabla_x^2 f(x_t) (x_{t+1} - x_t)$$

$$x_{t+1} = x_t - \nabla_x^2 f(x_t)^{-1} (\nabla_x f(x_t))$$

Properties

① $\nabla_x^2 f(x_t)^{-1} (\nabla_x f(x_t))$ is a newton direction.

$$x_{t+1} = x_t - u_t$$

If the funcⁿ decreases by making a movement towards u_t . We can hope for that to happen if

$$\langle \nabla_x f(x), u \rangle < 0$$

$$\langle \nabla_x f(x), -\nabla_x^2 f(x)^{-1} \nabla_x f(x) \rangle$$

$\nabla_x^2 f$ if it is sym. positive definite
all eigenvalues are > 0

$$(\nabla_x^2 f)^{-1} = U \Lambda^{-1} U^T$$

II Hessian inverse compensates for the difference in curvature.

$$f(x) = \frac{1}{2} x^T H x + g^T x + c$$

$$\nabla_x f(x) = Hx + g$$

$$\nabla_x^2 f(x) = H$$

$$x_{t+1} = x_t - \alpha (Hx_t + g)$$

$$x_{t+1} = x_t - \alpha H^{-1} (Hx_t + g)$$

$$= x_t - \alpha (x_t + H^{-1}g)$$

$$= x_t - \alpha (x_t - x_*)$$

→ GD

→ incremented
step in
newton
direcⁿ

$$x_{t+1} - x_* = x_t - x_* - \alpha (x_t - x_*)$$

$$= (1 - \alpha) (x_t - x_*)$$